The Langstroth bee hive is made up of wood and consists of six parts (Figure 12.6) namely **Stand, Bottom board, Brood chamber, Super, Inner cover** and **Top cover.** Besides the above primary equipments, other accessory equipments are used in beekeeping. They are **Queen Excluder, Comb foundation, Bee gloves, Bee veil**, **Smoker, Hive Tool, Uncapping knife, Bee brush, Queen introducing cage, Feeder, Honey Extractor** and **Hive Entrance Guard .**

Products of bee keeping and their economic importance

The chief products of bee keeping industry are honey and bee wax.

Honey is the healthier substitute for sugar. The major constituents of honey are: levulose, dextrose, maltose, other sugars, enzymes, pigments, ash and water. It is an aromatic sweet material derived from nectar of plants. It is a natural food, the smell and taste depends upon the pollen taken by the honey bee. It is used as an antiseptic, laxative and as a sedative. It is generally used in Ayurvedic and Unani systems of medicine. It is also used in the preparation of cakes, breads and biscuits

Bee wax is secreted by the abdomen of the worker bees at the age of two weeks. The wax is masticated and mixed with the secretions of the cephalic glands to convert it into a plastic resinous substance. The resinous chemical substance present in the wax is called **propolis** which is derived from pollen grains. The pure wax is white in colour and the yellow colour is due to the presence of carotenoid pigments. It is used for making candles, water proofing materials, polishes for floors, furniture, appliances, leather and taps. It is also used for the production of comb foundation sheets in bee keeping and used in pharmaceutical industries.



Bees teach us a lesson to work with cooperation. Imagine the hardwork of the bees! A single honey bee travels about double the distance of the circumference of the earth's globe for preparing 453.5ml of honey.

12.5 Lac Culture

The culture of lac insect using techniques for the procurement of lac on large scale is known as Lac culture. Lac is produced by the lac insect *Tachardia lacca* previously known as *Laccifer lacca.* It is a minute, resinous crawling scale insect which inserts its probosics into the plant tissues and sucks juice, grows and secretes lac from the hind end of the body as a protective covering for its body. Moreover the insect is a parasite on host plants i.e., Karanagalli (*Acacia catechu*), Karuvelai (*Acacia nilotica*) and Kumbadiri (*Schleichera oleosa*). The quality of lac depends upon the quality of the host plant. The female lac insect is responsible for large scale production of lac, which is larger than the male lac insect.

Economic importance of Lac

a. Lac is largely used as a sealing wax and adhesive for optical instruments. It is used in electric industry, as it is a good insulator.
b. It is used in preparations of shoe and leather polishes and as a protective coating of wood.

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